Lecture 14:

VLSI Data Phase Detector Circuits

Contents

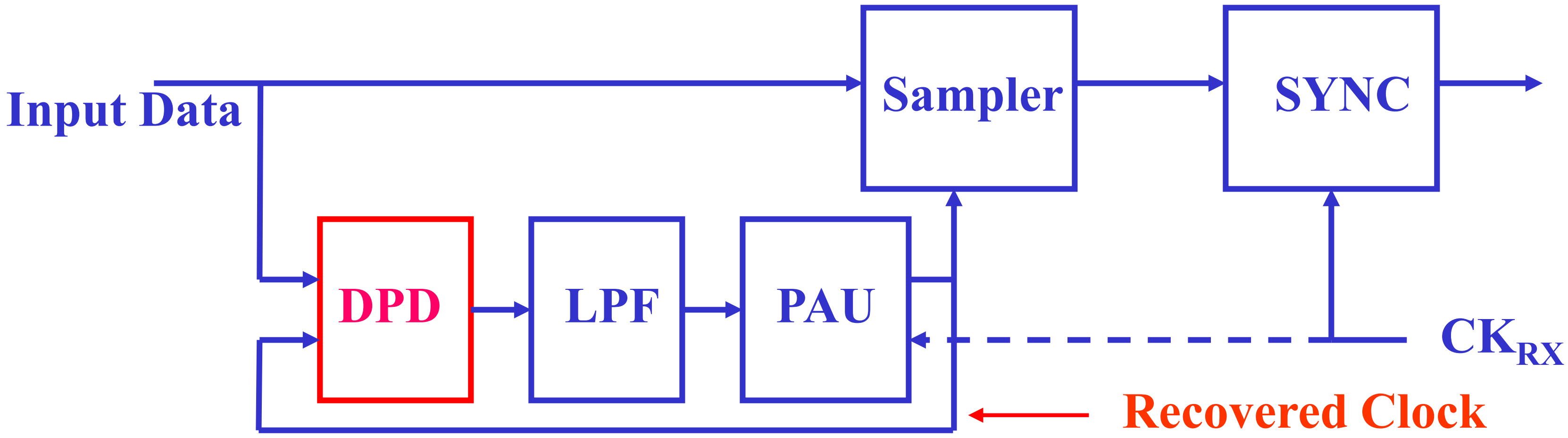
This Topic: VLSI Data Phase Detector (DPD) Circuits

Introduction

- VLSI DPD circuits are required to compare two TD signals with one as non-periodic (random) data signal
- The output of DPD circuit is a PWM or PCM signal
- The average of the DPD outputs over the one or multiple reference clock period is desired to be proportional to the delay (or phase) difference of the two TD signals
- VLSI DPD circuits are critical circuits in high-speed I/O DRC, DFT circuits

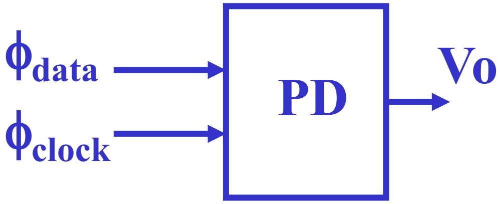
VLSI DPD Circuit Application: Rx

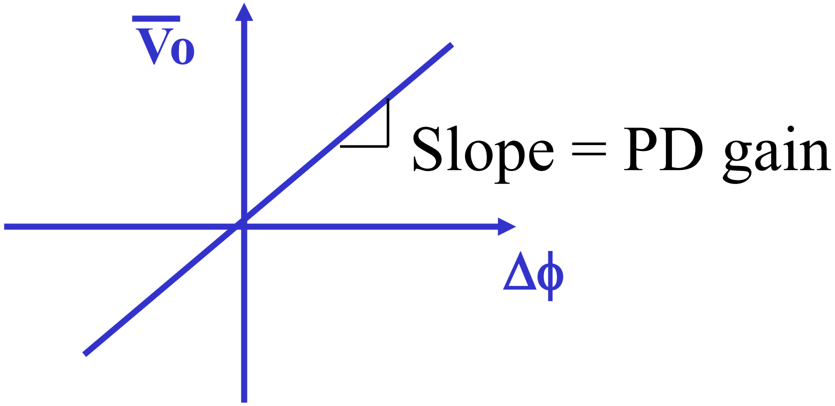
- Data Phase Detector (DPD)
- Loop Filter (LPF)
- Phase Adjustment Unit (PAU) (VCDL, VCO, PI)
- Rx Sampler
- Re-Timer (SYNC)



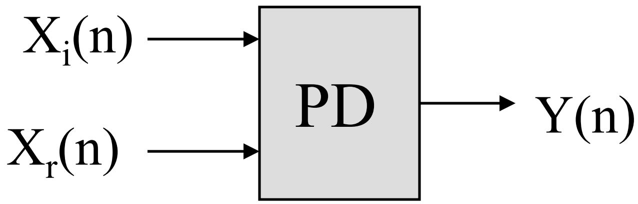
Basic VLSI DPD Operation

- Detect/measure the phase between the received data and Rx clock reference
- Provide a digital or analog <u>voltage</u> signal with its <u>average</u> proportional to the above phase difference
  - Similar to PLL, it need to handle frequency difference
  - Different from PLL, it also needs to handle transition density issue

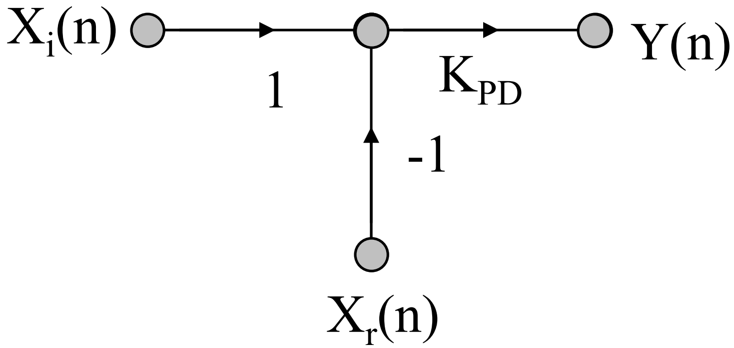


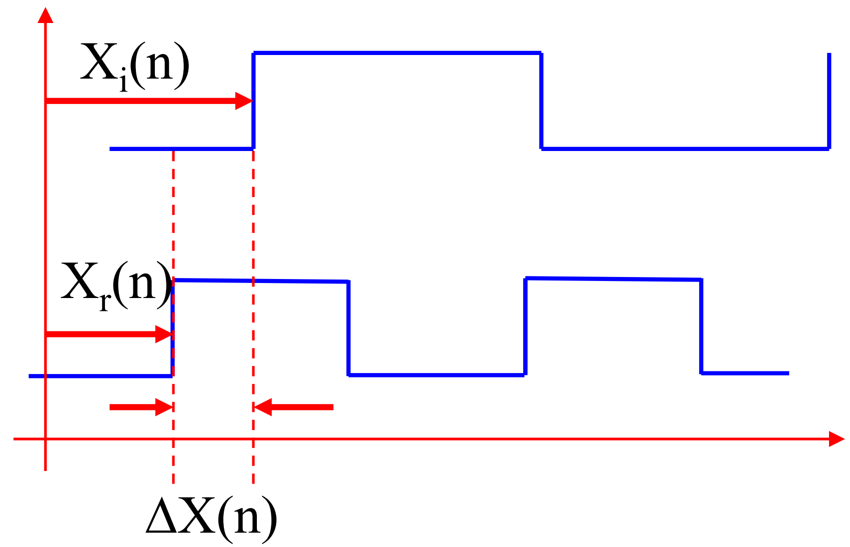


TDSP Model of DPD



$$\overline{Y(n)} = K_{PD}(X_i(n) - X_r(n))$$



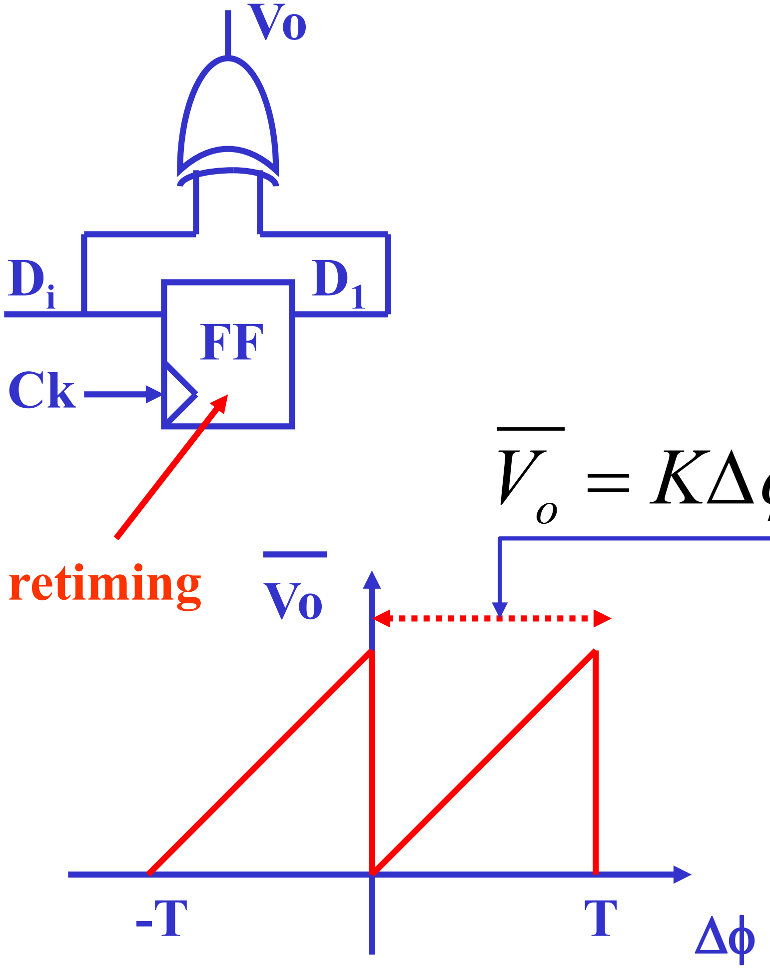


VLSI DPD Circuit Architectures

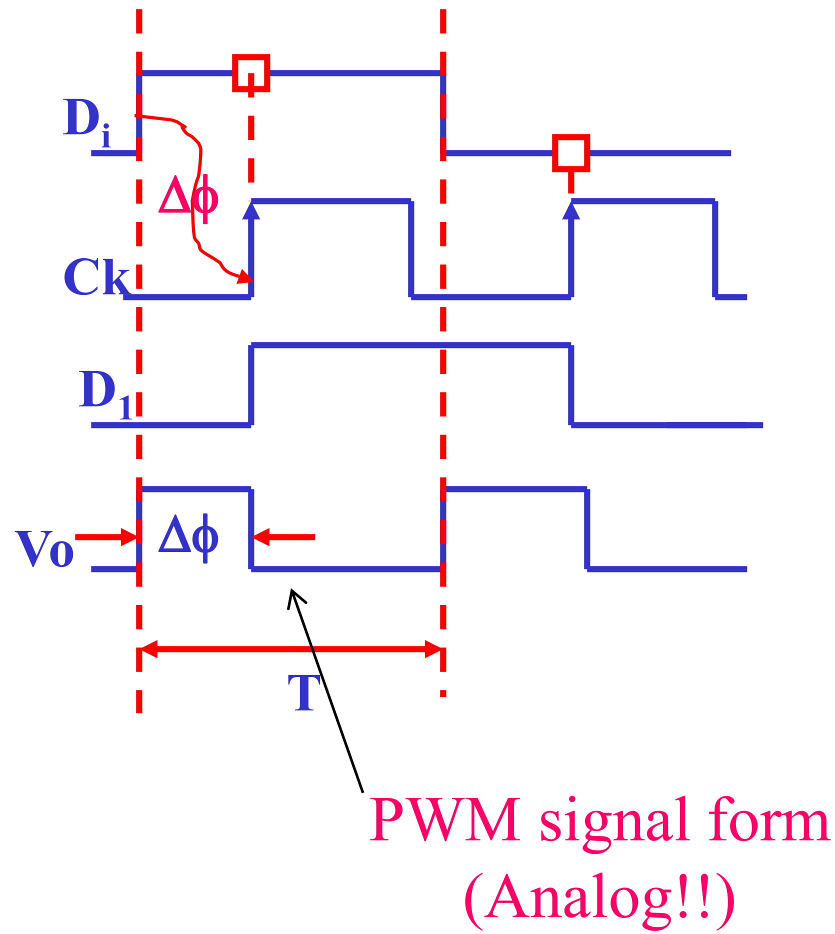
A Few VLSI DPD Circuit families

- Analog DPD Circuits
- Digital DPD Circuits
- Over-Sampling DPD Circuits
- Other DPD

Basic VLSI Analog DPD Circuit

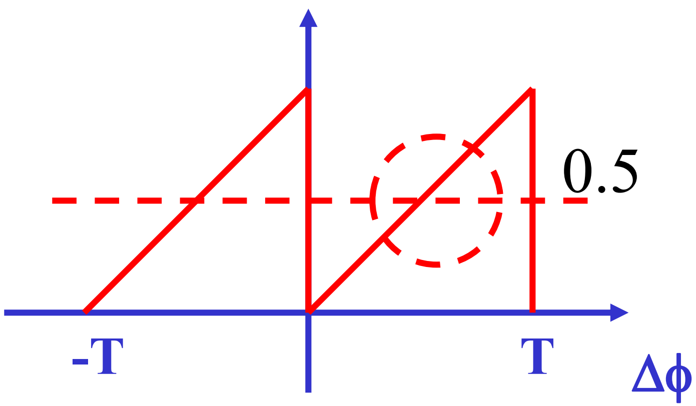


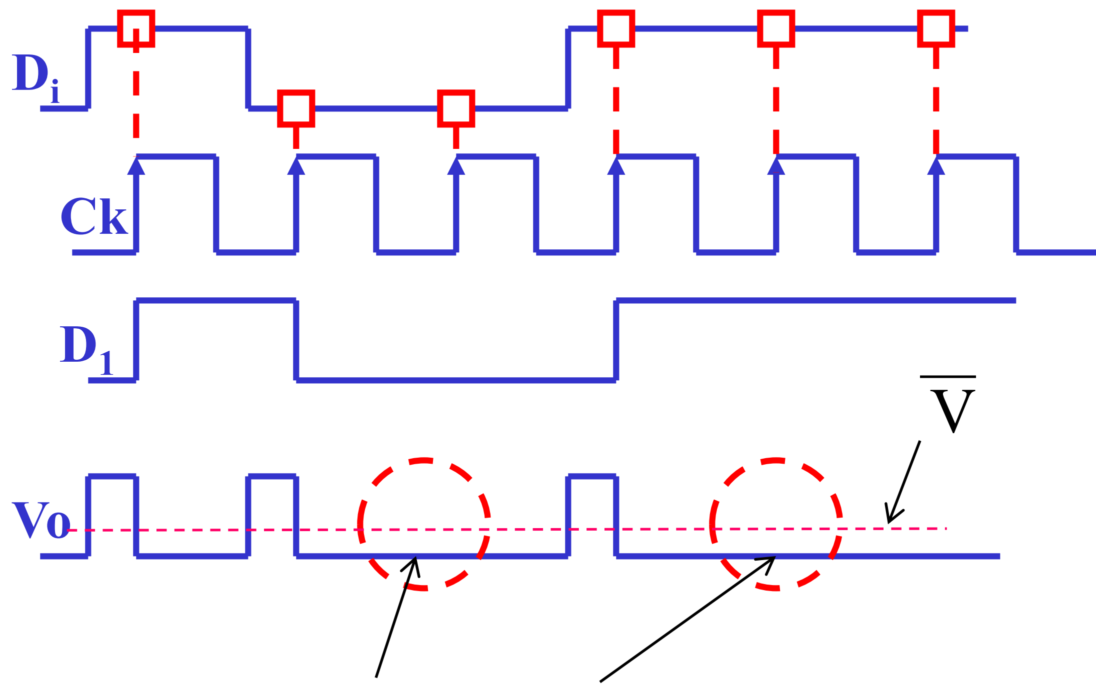
$$\overline{V_o} = K\Delta\phi$$



Issues in Basic Analog DPD

- DC offset issue due to
  - Pattern Sensitivity
  - Reference

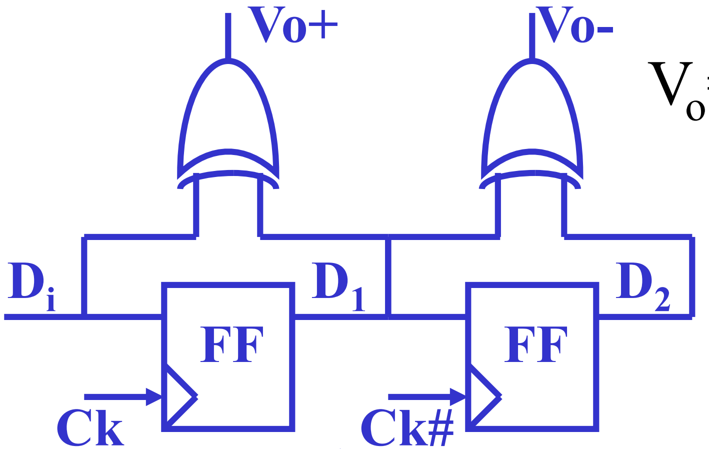




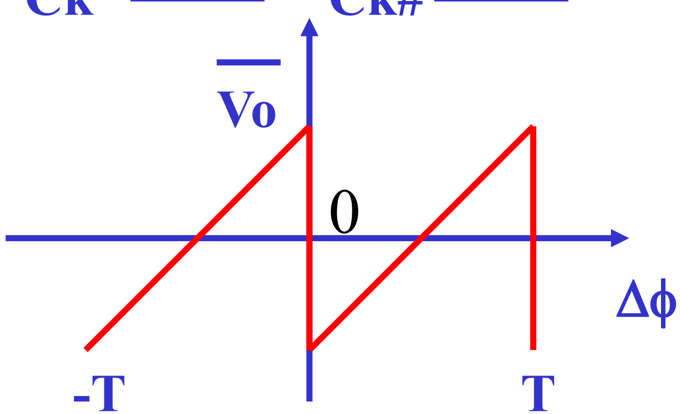
These missing transition will impact the averaging!!

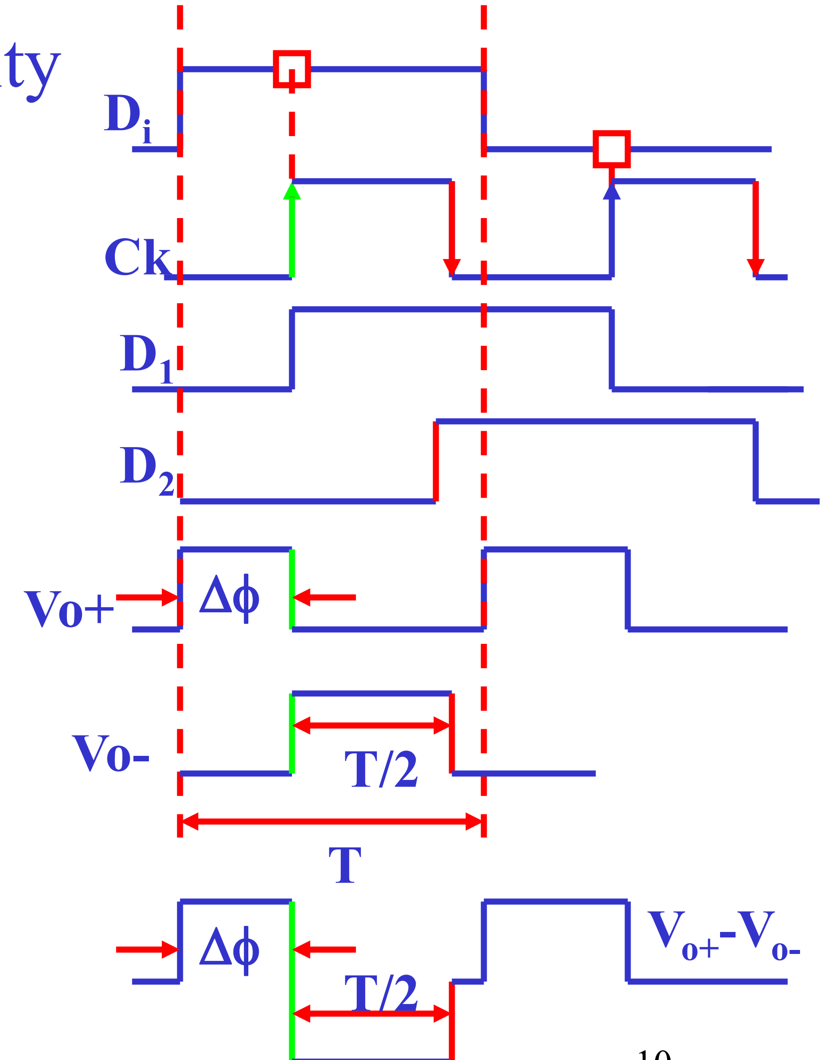
Improvement 1: Pseudo Differential Analog DPD Circuit

- Eliminated pattern sensitivity
  - Still sensitive to clock DCD



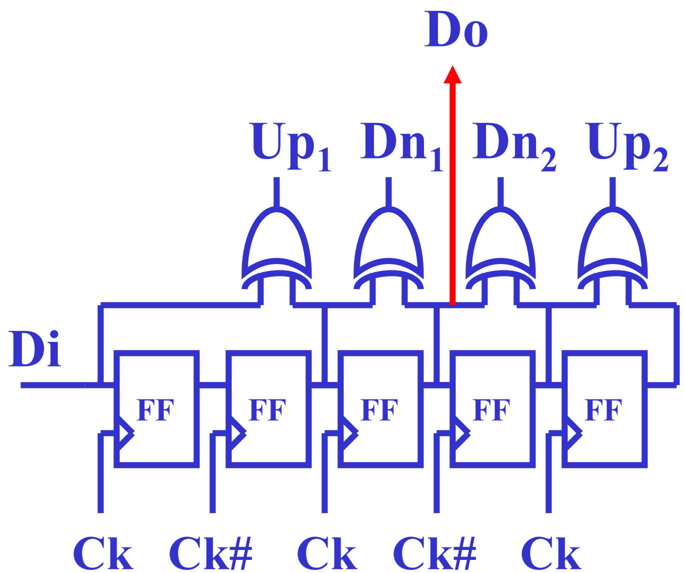
$$V_o = V_{o+} - V_{o-}$$

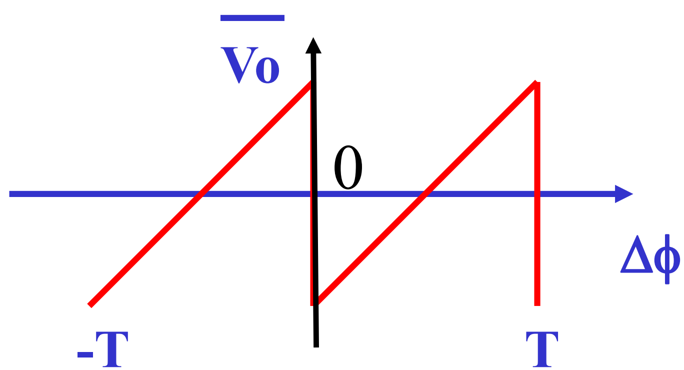


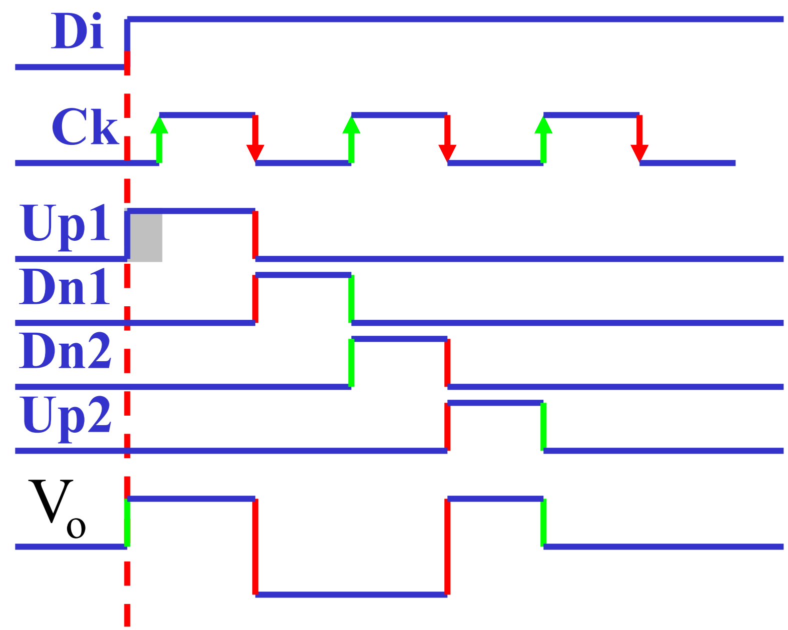


Improvement 2: Double Pseudo Differential Analog DPD Circuit

- Eliminated Pattern Sensitivity & DCD Sensitivity



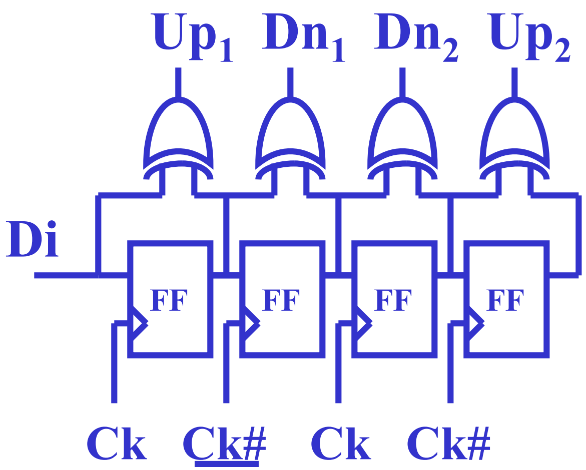


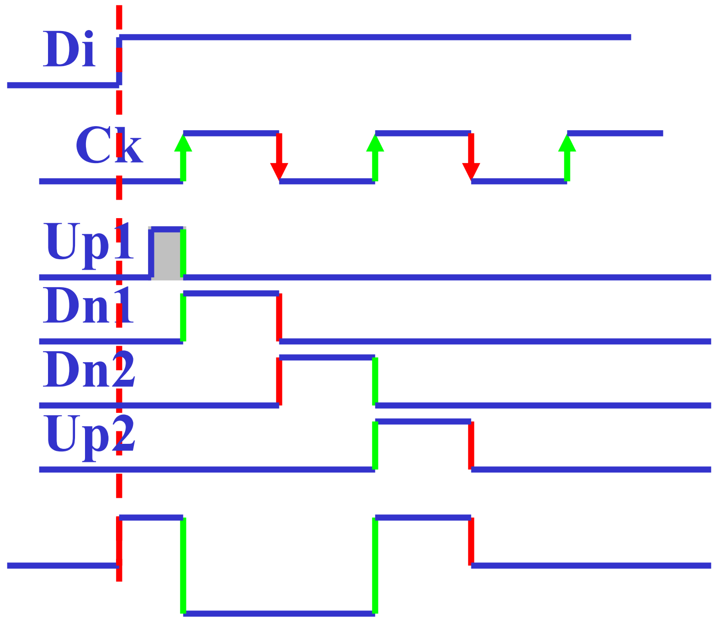


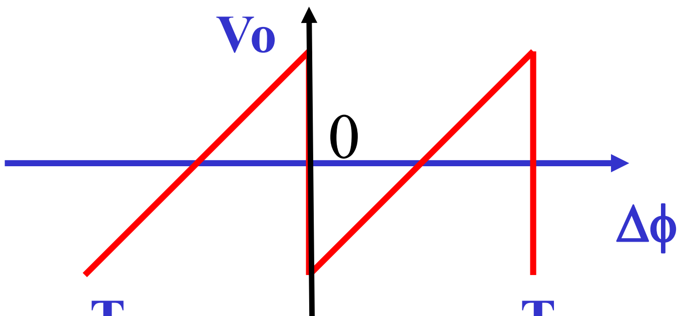
$$V_o = Up_1+Up_2-Dn_1-Dn_2$$

Alternative Double Pseudo Differential DPD

- Eliminated Pattern Sensitivity & DCD Sensitivity
- Reduced device count



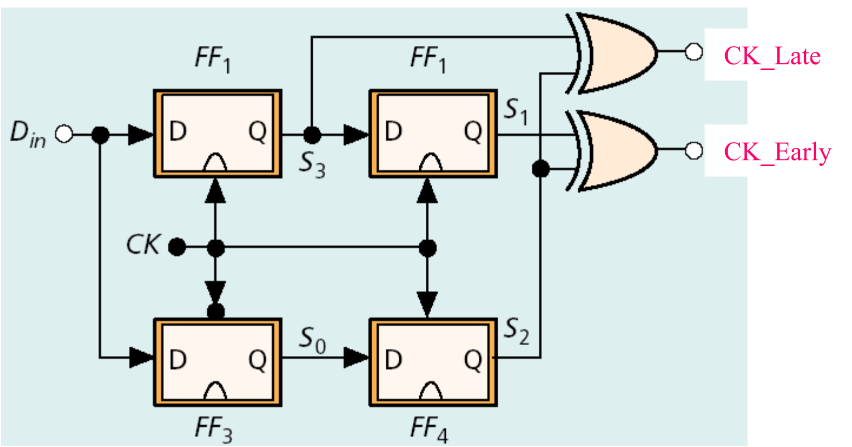


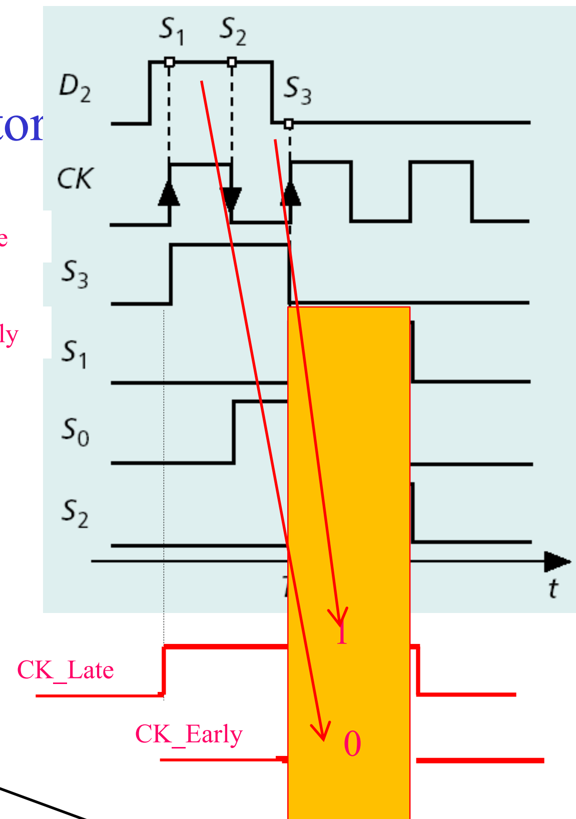


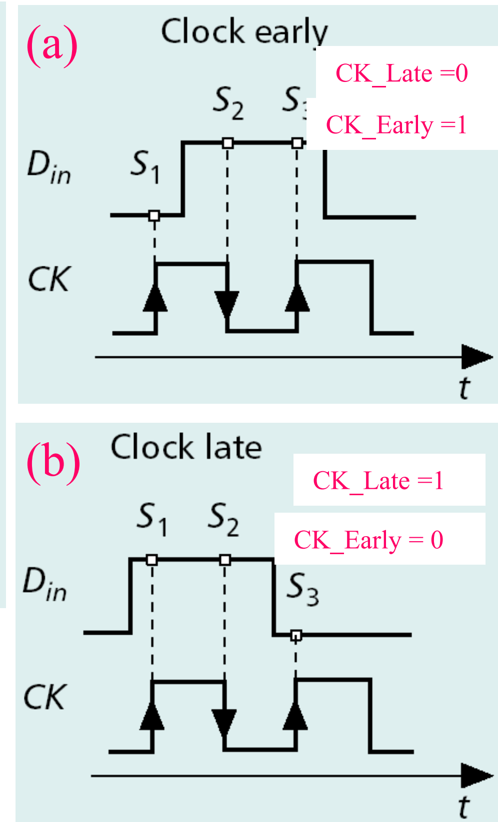
$$V_o = Up_1 + Up_2 - Dn_1 - Dn_2$$

Digital DPD Circuits

- Output of Digital DPD is a <u>digital signal</u>
  - (i.e. PCM form)
  - Alexander Phase Detector







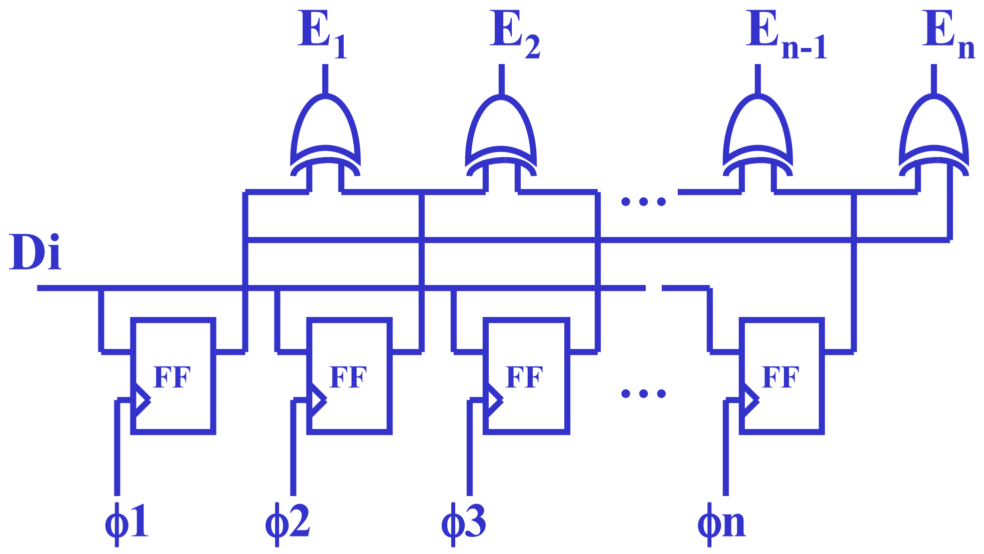
$$CK\_Late = S2 \oplus S3$$

$$CK\_Early = S1 \oplus S2$$

In PCM signal form (Digital!!)

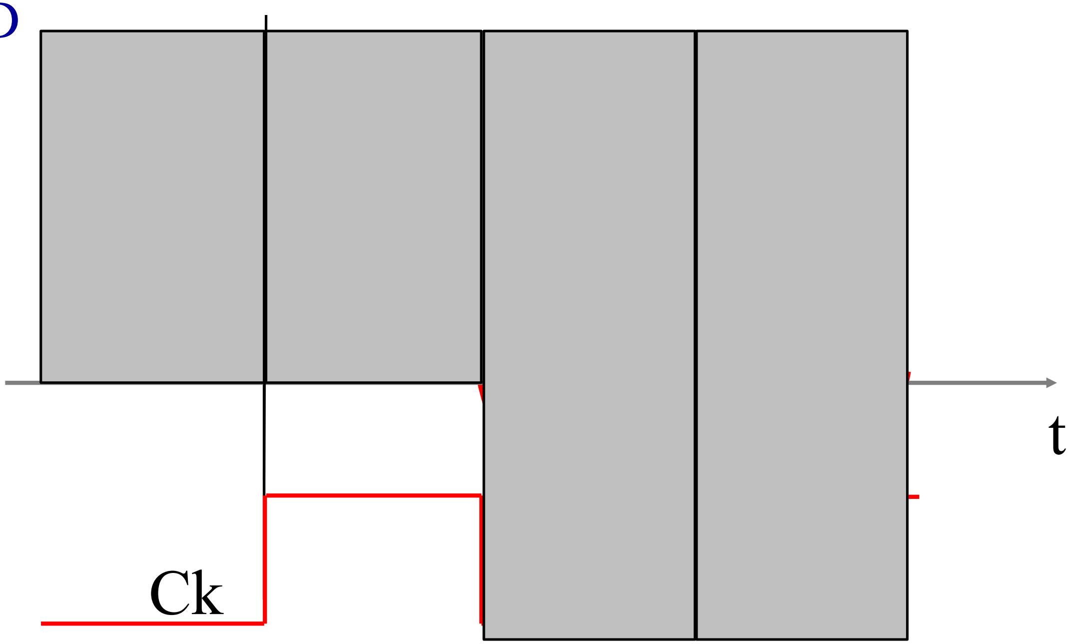
Over-Sampling DPD

- Realize a Phase-to-Digital Conversion
- Using a polyphase clock set



Other DPDs

- Non-data transition based DPDs
  - e.g. M-M DPD



CK_Lead = (V>0 & dV/dt>0) || (V<0 & dV/dt <0)

CK_Late = (V>0 & dV/dt<0) || (V<0 & dV/dt >0)